

MSc Health Data Science Project Title

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Background and Problem

Daily travel to school plays an important role in shaping physical activity levels among children and adolescents. In recent years, increased reliance on unhealthy modes of transport, such as private cars, has reduced opportunities for active travel, including walking and cycling. In Ireland, factors such as school location, travel distance, and neighbourhood characteristics strongly influence whether students use healthy or unhealthy ways of travelling to school.

Reduced physical activity linked to unhealthy transport choices contributes to rising levels of overweight and obesity in the population. Encouraging active school travel is therefore a key public health concern, particularly as travel habits formed during childhood often persist into adulthood.

Objectives of Project

1. Map and analyse the spatial distribution of schools across Ireland.
2. Examine the distance between students’ residential locations and their schools, and investigate student mobility related to school attendance.
3. Analyse school travel behaviour and assess the prevalence of healthy versus less healthy commuting modes.
4. Explore the relationship between school proximity, travel mode, and potential health implications.

Data Sources and Datasets

The syntax in this poster template and the `posterdown` package uses the same workflow

approach as the R Markdown you know and love.

You can even use the bibliography the same way: Our data were taken from a cluster-randomised trial¹, available from the [Irish Social Science Data Archive](#).

Early Results / Descriptive Statistics of Datasets

Usually you want to have a nice table displaying some important results that you have calculated. In `posterdown` this is as easy as using the `kable` table formatting you are probably use to as per typical R Markdown formatting.

You can reference tables like so: Table 1. Some basic summaries of the dataset are below:

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
5.1	3.5	1.4	0.2
4.9	3.0	1.4	0.2
4.7	3.2	1.3	0.2
4.6	3.1	1.5	0.2
5.0	3.6	1.4	0.2
5.4	3.9	1.7	0.4
4.6	3.4	1.4	0.3
5.0	3.4	1.5	0.2
4.4	2.9	1.4	0.2
4.9	3.1	1.5	0.1

Figure 1, and Figure 2 below show the patterns in our dataset. Make sure that all the details in your plots will be legible when printed (legend text, axis text, and any labels)

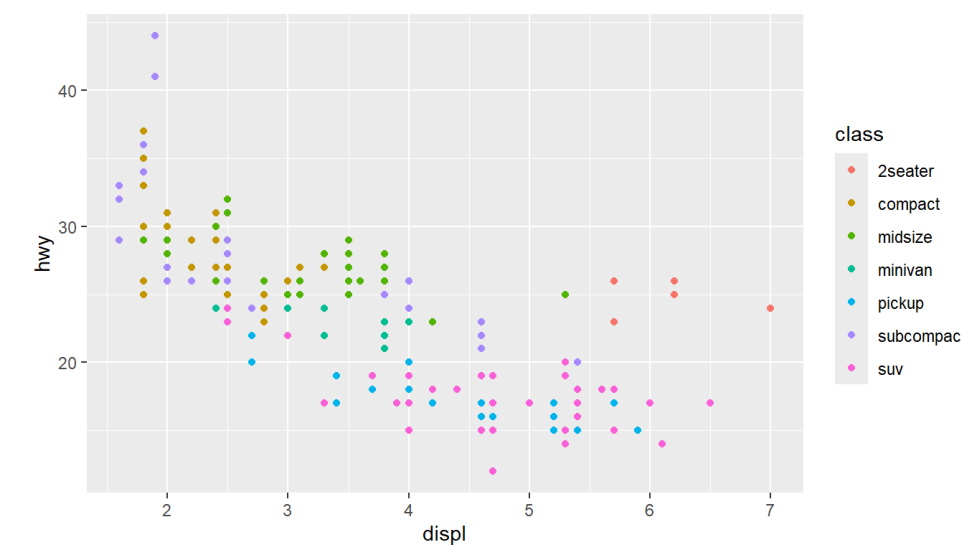


Figure 1: Great figure!

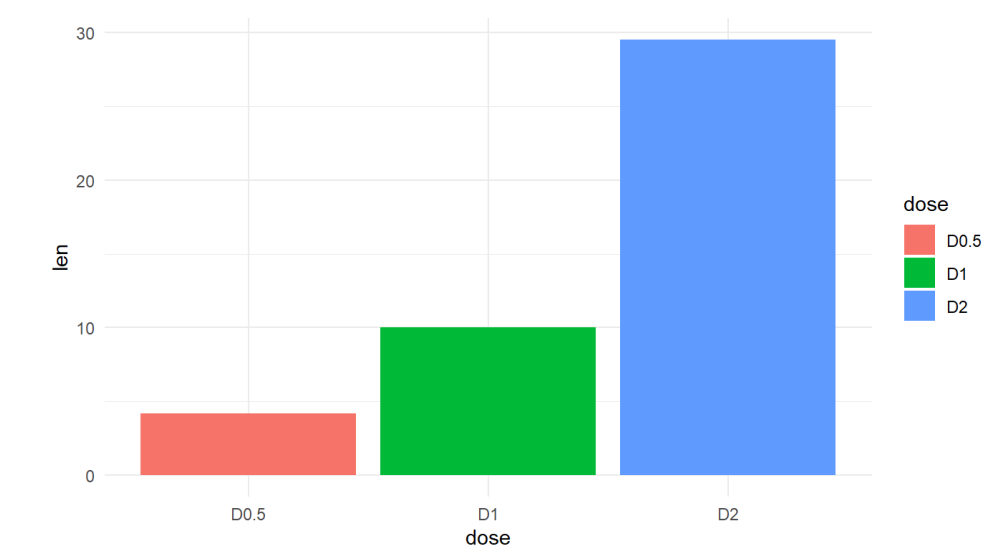


Figure 2: Amazing, right?!

You can even make your plots interactive for the HTML version of the poster. You can use the HTML poster for the presentation session, and the PDF poster will be printed - so be sure the static version looks okay.

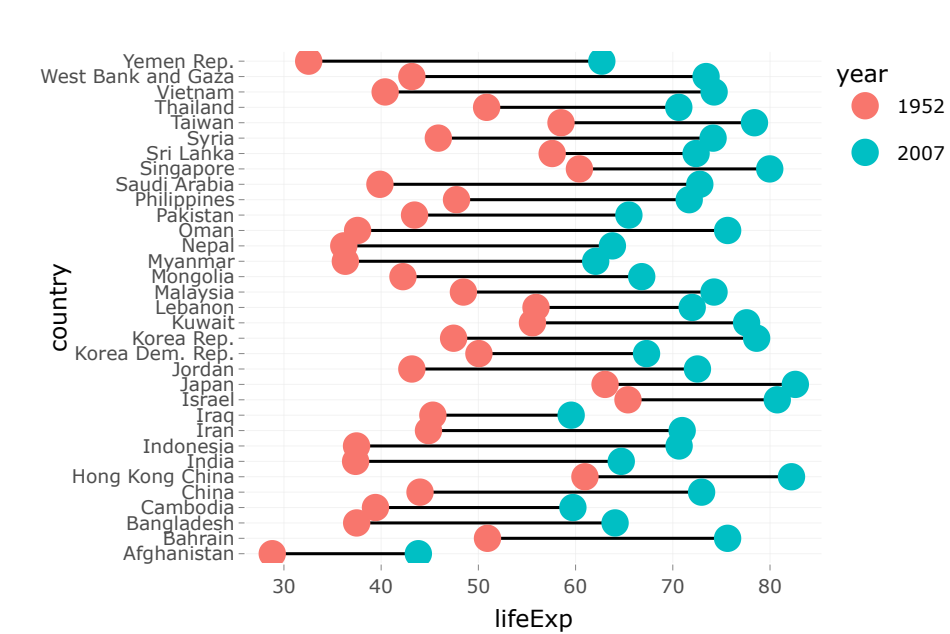


Figure 3: Amazing, right?!

Next Project Steps

We plan to conduct further analysis using:

- Variable discombobulation²
- Expand our minds with explosive machine learning³.

We will use the `plasticanalysis` package for this.

GitHub

The code and datasets for this project can be viewed at our GitHub repository here: <https://github.com/judithbeltranlopez/Final-Project>

References

1. Murphy et al. 2005 doi: 10.1186/1468-6708-6-11↗
2. Massey et al. 2005 doi: 15.36.413↗
3. Smith et al. 1991 doi: 12.36.486↗